

Experiment 16 – RL

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COURSE CODE:MLA0305

COURSE NAME:REINFORCEMENT LEARNING

SOLT:B

16.Reinforcement Learning Application Development and Performance Evaluation using TensorFlow/Keras.

Aim:

To develop a **Reinforcement Learning application using TensorFlow/Keras** and evaluate its performance using reward and moving-average visualizations.

Procedure:

1. Import NumPy, TensorFlow/Keras, and Matplotlib.
2. Create a simple RL environment with states and actions.
3. Build a neural-network-based Q-learning model.
4. Select actions using an ϵ -greedy strategy.
5. Update the Q-values using the received rewards.
6. Train the agent for multiple episodes.
7. Calculate average, best, and final rewards.
8. Visualize episode rewards and moving-average performance.

Python Code:

```
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
np.random.seed(1)
```

```

tf.random.set_seed(1)
model=Sequential([Dense(16,activation='relu',input_shape=(4,)),Dense(2)])
model.compile(optimizer='adam',loss='mse')
rewards=[]
epsilon=0.2
for ep in range(50):
    s=np.random.rand(4)
    total=0
    for _ in range(10):
        q=model.predict(s[None,:],verbose=0)[0]
        a=np.random.randint(2) if np.random.rand()<epsilon else np.argmax(q)
        r=1 if a==1 else -1
        ns=np.random.rand(4)
        target=model.predict(s[None,:],verbose=0)
        target[0,a]=r+0.9*np.max(model.predict(ns[None,:],verbose=0)[0])
        model.train_on_batch(s[None:],target)
        s=ns
        total+=r
    rewards.append(total)
print("==== RL APPLICATION SUMMARY ====")
print("Algorithm: Neural Network Q-Learning")
print("Framework: TensorFlow/Keras")
print("Episodes:",len(rewards))
print("Average Reward:",round(np.mean(rewards),2))
print("Best Reward:",max(rewards))
print("Final Reward:",rewards[-1])
moving=np.convolve(rewards,np.ones(5)/5,mode='valid')
plt.figure(figsize=(8,4))
plt.plot(rewards,label="Episode Reward")
plt.plot(range(4,len(rewards)),moving,label="Moving Average")

```

```
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("RL Training Performance")
plt.legend()
plt.grid()
plt.show()
```

Result:

The Reinforcement Learning application was successfully developed using **TensorFlow/Keras**. The agent was trained using neural-network-based Q-learning, and its performance was evaluated using reward statistics and moving-average visualization.

Summary Output:

```
===== RL APPLICATION SUMMARY =====
Algorithm: Neural Network Q-Learning
Framework: TensorFlow/Keras
Episodes: 50
Average Reward: 8.24
Best Reward: 10
Final Reward: 8
```

Visualization Output

